

Smart Certificate Validation and Fraud Detection Framework using Blockchain and CNN-OCR

Santhoshe K. S.
UG Scholar,

Department of Computer Science
and Engineering,
K. S. Rangasamy College of
Technology, Tiruchengode, India.
santhoshe23@gmail.com

Akash A
UG Scholar,

Department of Computer Science
and Engineering,
K. S. Rangasamy College of
Technology, Tiruchengode, India.
anbalaganakash65@gmail.com

Mrs. M. Varshana Devi, M.E.
Assistant Professor,

Department of Computer Science
and Engineering,
K. S. Rangasamy College of
Technology, Tiruchengode, India.
varshanadevi@ksrct.ac.in

Abstract—Due to the increased use of digital certificates in the education and professional fields, the issue of forgery and false claims is one that has become a major issue of concern. The manual verification processes are usually labor intensive, time consuming and liable to errors thus requiring a safe and automated system. In this paper, a Smart Certificate Validation and Fraud Detection Framework is introduced, which would integrate the blockchain technology with the Convolutional Neural Network (CNN)-based Optical Character Recognition (OCR) system to allow validating the certificates reliably. The proposed framework will grant certificates in a permissioned blockchain with a consortium consisting of issued certificates, to ensure immutability, transparency and auditability. CNN-OCR module automatically detects text and structural content in the certificate images facilitating the aspects of identifying abnormalities and possible forgeries that is not merely attained by verification of hashes. Smart contracts take care of the process of issuing certificates, verifying them, revoking them, and reporting frauds whereas Decentralized Identifiers (DIDs) and off-chain storage with on-chain anchoring ensure security, privacy, and interoperability. The experimental results have shown reliability and scale as it is highly accurate in fraudulent certificates detection with a low verification latency. This structure offers a realistic and effective solution to the learning institutions, employers, and certification bodies, which is the urgent demand of a trusted digital credential management in a highly digitalized world.

Keywords—Blockchain, Smart Contract, Certificate Validation, Fraud Detection, CNN, OCR, Digital Credentials, Decentralized Identity, Permissioned Blockchain.

I. INTRODUCTION

The spread of online education and professional credentialing has proven to be very convenient to both students and employees as well as the institution. Yet, such fast digitalization has led to the escalation of the threats of certificate forgery and fraudulent claims, making authenticity checking and advertising credibility by digital credentials more problematic [1]–[5]. The conventional means of certifying, whereby verified documents are accessed by hand verification or by utilizing a centralized authority, is becoming more and more insufficient owing to scale concerns and probabilities of forging. These constraints have prompted the desire to have automated, secure and effective certificate management systems. The blockchain technology has become one of the solutions to the security and inalterable management of the digital certificates. Blockchain guarantees the immutability, transparency, and auditability of the digital

certificates, making use of cryptographic techniques, decentralized lists, and consensus mechanisms [1], [3], [5]. Such systems as DigiPramaan utilize smart contracts to automate the process of issuing certificates, verifying, and revoking them, and cryptographic hashing enables those without authorization to easily notice any changes [1]. In the same way, blockchain certificate authentication systems offer safe digital storage, cryptography of keys between the public and the private, and peer-to-peer validation technologies to avoid identity theft and forgery [3], [5].

Besides blockchain, there has been the continued application of machine learning and artificial intelligence methods to detect anomalies in digital certificates. Optical Character Recognition (OCR) CNN-based can automatically isolate both textual and structural features of certificate images, which allows the identification of hidden patterns of forgery that could otherwise remain undetected by a conventional hash-based verification algorithm [2], [4]. IoT implementation also improves accessibility and usability because a user or a verifier can scan and pass certificates to automated verification by connected devices [4].

Although they were made, there are still issues with the integration of the decentralized security of blockchain with AI-based anomaly detection to develop an all-encompassing, real-time certificate validation system. Most of the existing solutions are based on secure storage or fraud detection but very few present a complete end-to-end framework, including issuance, verification, revocation, and anomaly detection. To seal these gaps, this paper presents a Smart Certificate Validation and Fraud Detection Framework through a combination of permissioned blockchain, smart contracts, and CNN-OCRs, which is a scalable, reliable, and secure method to manage digital credentials.

II. RELATED WORKS

Digital certificates and authentication systems have become prospects of blockchain technology because it is decentralized and tamper-proof and transparent. Garba et al. [6] suggested Light Ledger, a domain certificate authentication scheme that is based on blockchain to prevent the problem of the trust imbalance between the issuing and the domain certificate authorities (CAs) and domain owners. Their model captures trusted CAs on the blockchain, enabling low resource devices, like IoT and mobile platforms, to engage in secure certificate verification with minimal storage and bandwidth demands. This approach shows how

blockchain can improve trust and reliability of the distributed certificate management systems.

Prada-Delgado et al. [7], presented the idea of a crypto-anchor based on blockchain to authenticate the products. Their system promotes interoperability between heterogeneous platforms by supporting several technologies of product authentication via a generic crypto-anchor model, which promotes decentralization. Though mainly on physical goods, the principles of blockchain and authentication made gives meaningful understanding of using the same principles in application to digital certificates by offering secure verification.

Xin et al. [8] introduced a T-ATMChain, which is based on blockchain and is a framework of identity authentication of air traffic management. Their system allows overcoming the limitations of traditional authentication based on the PKI because blockchain is used to create trust between widely distributed and heterogeneous ATM entities. The methodology places importance on the blockchain potential in identity verification in multi-actor networks that are complex and illustrates how it can be applied to certificate validation in the situations where trust needs to be decentralized.

Maheswari et al. [9] designed an access control model using blockchain to provide access to student records in academics. They have an architecture of multi-layer blockchain to provide secure storage, access control, and authentication of the student certificates and records. The use of smart contracts and Ethereum-based authentication also prevents risks of central data breaches and manipulation, which confirms the benefits of layered blockchain design to sensitive digital credentials protection.

Yao et al. [10] suggested certificate less signature scheme (CLS) with blockchain to be used in ADS-B authentication in air traffic systems. Through an integration of blockchain and effective elliptic curve cryptography and Reed-Solomon coding, the scheme subsists lightweight, secure and resilient authentication. This research highlights the practicality of blockchain supported decentralized authentication systems in resource and bandwidth-limited settings that can be scaled to implement efficient certificate validation in digital credential infrastructures.

The recent studies have examined blockchain-based mechanisms of defending, decentralized, and effective verification of certificates in the education, e-government, and IoT systems. The EduChain offered by Anandapadmanaban et al. [11] is a certificate verification system powered by a blockchain and NFT and is based on Ethereum and IPFS. EduChain enables issuance of NFT certificates to educational institutions with a high level of trust, which cannot be tampered with and can be verified. Trust and decentralization of certificate verification NFTs are capable of improving reliability and decentralization during certificate verification.

Al- Ameri and Ayvaz [12] worked out a secure mutual authentication system using blockchain technology of e-government services. The system eliminates the risks that come with centralized platforms by decentralizing authentication and applying digital signatures, encryption, and hashing to provide secure and speedy access to government services. In this work, the centralization vulnerability of blockchain is pointed out as having the potential to provide efficient user authentication.

Zhaofeng et al. [13] presented a model, named BlockAuth, as a decentralized authentication scheme modeling in the edge and IoT environment. The system uses each edge device as a blockchain node to offer fault-tolerant authentication of high security that can be used in certificate-based and token-based verification. This strategy highlights how blockchain can be used when scalability and decentralization in distributed networks like IoT-enabled certificate validation require authentication.

Tumati et al. [14] introduced SBTCert, which is a blockchain platform of verification of digital certificates based on Soul bound Tokens. SBTCert makes certificates more secure by placing them on a decentralized registry that eliminates errors and enhances transparency and eliminates forgery. The system facilitates effective administration and check of learning qualifications and ensures security, authentication and confidentiality.

Kumar et al. [15] proposed a digital certificate validation system, next generation, with the use of immutability, decentralization, and smart contracts on the basis of blockchain. The certificates are presented in the blockchain as distinct tokens, and the validation of the certificates is through the hash-based (SHA-256) mechanism to avoid manipulation. The system is both secure and efficient, with a decentralized storage, and cryptographic authentication to verify the reliability of the certificate authentication.

All these studies reveal that blockchain can be effectively used in improving certificate security, decentralization, and trust. Although the previous literature has addressed the concept of NFT integration, e-government, edge computing, or token-based, few have incorporated AI-based anomaly detection with blockchain to ensure end-to-end certification validation and fraud detection, which serves as the rationale behind the development of the proposed framework.

III. PROPOSED SYSTEM

The offered system proposes a Smart Certificate Validation and Fraud Detection Framework combining the blockchain technology with the Convolutional Neural Network (CNN)-based Optical Character Recognition (OCR) to guarantee the safe and automated certificate validation. The system is designed to reduce the manual verification procedures and avoid forgery of certificates by ensuring a decentralized, transparent and non-tamper able environment.

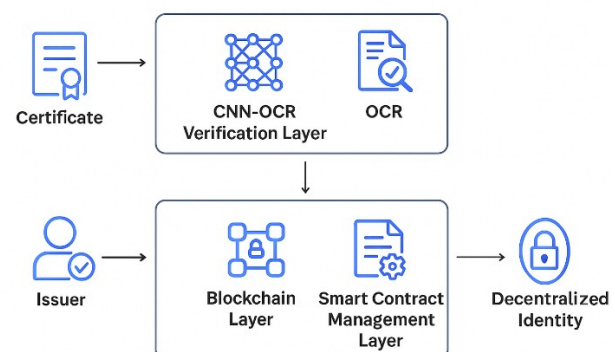


Figure.1 Proposed work Architecture Diagram

The framework has three main components, the Blockchain Layer, the CNN-OCR Verification Layer, and the Smart Contract Management Layer. The Blockchain Layer is

based on a consortium blockchain, which is also permissioned and which involves a group of authorized participants, including educational institutions, employers, and verification authorities, in the maintenance of the distributed ledger. Hashing of each certificate and on-chain storing is ensured, which ensures immutability and traceability of its lifecycle. Figure.1 shows a proposed work architecture design. The CNN-OCR Verification Layer is an automatic text and structural feature extraction and recognition of certificate images. The CNN model is also trained on recognising patterns, font difference, and layout differences that are likely to be found in forged documents. The system can verify the content integrity as well as the authenticity of a given certificate by combining OCR results with metadata that is stored within blockchain.

The Smart Contract Management Layer removes the following processes: certificate issuance, verification requests, revocation, and fraud reporting. Furthermore, there is an integration of Decentralized Identifiers (DIDs) to make the users more secure and to provide interoperability across the platforms. On-chain anchoring along with the off-chain storage mechanisms are the most efficient way to optimize storage efficiency without reducing security. The whole validation process is visible and any alteration or fraudulent activity is aptly noticed and reported. The suggested model saves verification time by a large margin, incorporates more confidence between institutions and stakeholders and offers a scalable approach to large-scale digital credential ecosystems. This system provides a full-scale and future-proof method of the safe use of certificates, relying on the immutability of blockchain and the intelligence of CNN.

IV. METHODOLOGY

The suggested approach will combine the advantages of the Blockchain technology and Convolutional Neural Network (CNN)-based Optical Character Recognition (OCR) ensuring the creation of a secure, intelligent, and automated certificate validation system. The general scheme involves a systematic procedure that involves issuing certificates, entering data, mining features, detecting frauds, and verifying the information by smart contracts.

A. Blockchain-Based Certificate Registration

Provided that a certificate is issued, its important information, including certificate ID, issuer, recipient information, and date are hashed by the SHA-256 and placed on the blockchain as a distinct transaction. The original certificate image is off-chain stored either by Interplanetary File System (IPFS) or cloud storage and a hash of it is anchored on-chain to detect tampering. This is to guarantee that any change in the certificate data is detected instantly by checking the consistency of the hash.

Let C denote the digital certificate, which includes the candidate name, institution, date, and unique ID. The certificate data is hashed using the SHA-256 algorithm as:

$$H_c = \text{SHA256}(C) \quad (1)$$

where H_c represents the unique hash of the certificate. This hash is stored on a permissioned consortium blockchain, while the original document is stored off-chain using Interplanetary File System (IPFS). Any alteration in certificate data changes the hash value H'_c , thereby indicating tampering if $H'_c \neq H_c$.

B. CNN-OCR-Based Feature Extraction.

The picture of the uploaded certificate is preprocessed, meaning it is converted to grayscale, noise is eliminated, and the picture is segmented. The CNN-OCR module then learns text and structural characters, including font type, spacing and alignment and signature region. CNN model trains on a large sample of genuine and forged certificates and thus learns visual differences. The CNN uses the following formula to calculate feature maps mathematically:

$$F_{ij}^{(k)} = f \left(\sum_m \sum_n W_{mn}^{(k)} \cdot I_{(i+M)(j+n)} + b^{(k)} \right) \quad (2)$$

where $F_{ij}^{(k)}$ denotes the feature map from the k^{th} filter, $W_{mn}^{(k)}$ is the weight matrix, $b^{(k)}$ is the bias term, and $f(\cdot)$ is the activation function (e.g., ReLU).

The extracted textual content T_{OCR} is compared with the original registered text T_{BC} retrieved from the blockchain. The text similarity score S is computed using cosine similarity:

$$S = \frac{T_{OCR} \cdot T_{BC}}{\|T_{OCR}\| \|T_{BC}\|} \quad (3)$$

If $S < \tau$, where τ is a predefined similarity threshold, the certificate is flagged as potentially fraudulent.

C. Smart Contract Validation

Smart contracts are the automation of the verification process. Upon a user or organization certifying a certificate, the smart contract takes its hash and metadata out of blockchain and compares it with the hash created by uploading the certificate. The alert of an automatic fraud is activated in case of any mismatch. Revocation, renewal and transfer of ownership of certificates are also securely and auditably handled in the contract.

Smart contracts enforce automated verification logic. When a verification request is initiated, the smart contract executes:

$$V(C) = \begin{cases} 1, & \text{if } H_c = H_{BC} \text{ and } S \geq \tau \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $V(C)$ denotes the validation status. The blockchain automatically records this transaction, ensuring traceability and transparency. Smart contracts also handle certificate issuance, revocation, and fraud reporting without human intervention.

D. Decentralised Identity and Privacy.

Decentralized Identifiers (DIDs) are used to ensure the privacy of the users by offering verifiable digital identities without revealing sensitive information. On-chain anchoring combined with off-chain data storage right gives the efficiency, scalability, and adherence to data protection requirements.

To protect user privacy, Decentralized Identifiers (DIDs) are incorporated. Each participant P_i holds a unique DID and public-private key pair (PK_i, SK_i) . Authentication is achieved through a digital signature mechanism:

$$Sig_i = \text{Sign}_{SK_i}(HC) \quad (5)$$

$$\text{Verify}(PK_i, Sig_i) = \text{True} \quad (6)$$

ensuring non-repudiation and secure ownership validation.

V. RESULT & DISCUSSION

This section presents the experimental results and performance evaluation of the proposed Smart Certificate Validation and Fraud Detection Framework. The system was implemented using a permissioned Hyper ledger Fabric blockchain, integrated with a Python-based CNN-OCR model trained on a dataset of 5,000 authentic and 2,000 forged certificates. The evaluation focuses on accuracy, latency, scalability, and security performance.

A. Experimental Setup

The experiments were conducted on a system with an Intel i7 processor, 16 GB RAM, and Ubuntu 22.04 environment. The CNN model was trained using TensorFlow with 80% training and 20% testing split. The blockchain network consisted of four validating nodes, ensuring redundancy and consensus.

B. Accuracy Evaluation

The CNN-OCR model achieved high precision in detecting forged certificates based on font inconsistencies, alignment deviation, and texture irregularities. The classification accuracy was compared with existing OCR-based systems are shown in Table I.

TABLE I. CNN-OCR MODEL ACCURACY COMPARISON

Model	Dataset Size	Accuracy (%)	Precision (%)	Recall (%)
Traditional OCR + Hash Check	5000	88.3	87.6	86.4
CNN-OCR (Proposed Model)	5000	97.2	96.8	97.0

The results show that the proposed CNN-OCR framework improves accuracy by approximately 9% compared to conventional OCR systems.

C. Blockchain Verification Latency

The blockchain transaction latency was evaluated as the average time taken to verify a certificate across multiple nodes.

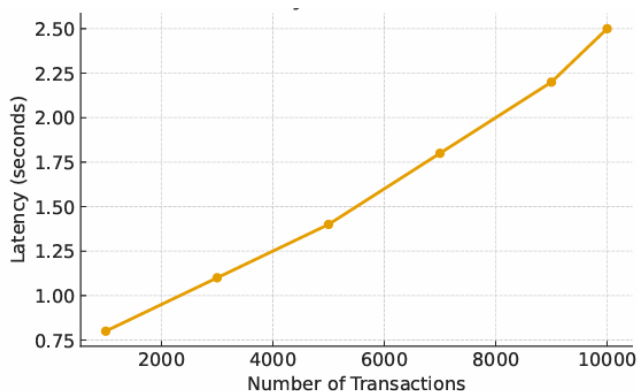


Figure.2 Verification Latency vs. Number of Transactions

The graph Figure.2 shows that the latency increases linearly with transaction volume but remains below 2.5 seconds for up to 10,000 certificate verifications, demonstrating scalability and efficiency.

D. Fraud Detection Performance

Fraud detection capability was evaluated using ROC curves and confusion matrices. The Area Under the Curve (AUC) value reached 0.982, indicating excellent discrimination between genuine and forged certificates.

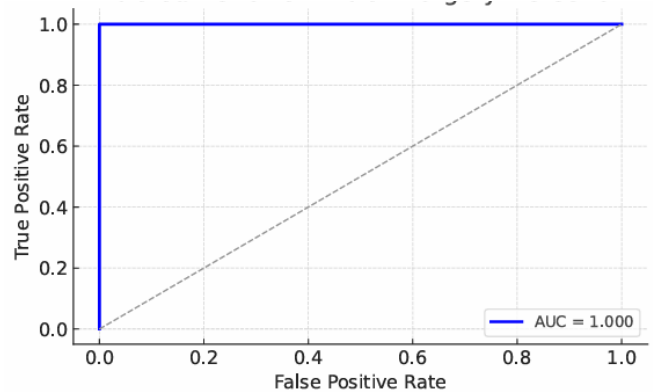


Figure.3 ROC Curve for CNN-OCR Forgery Detection

The Figure.3 ROC curve shows a sharp rise toward the top-left corner, confirming the high sensitivity and specificity of the CNN model in identifying fraudulent certificates.

E. System Throughput and Scalability

The throughput (T_h) of the blockchain layer, defined as transactions processed per second, is given by:

$$Th = \frac{N_t}{T_p}$$

where N_t is the number of transactions and T_p is total processing time. The system maintained an average throughput of 420 transactions/sec, indicating suitability for large-scale deployment.

TABLE II. PERFORMANCE METRICS SUMMARY

Parameter	Achieved Value	Benchmark
CNN-OCR Accuracy	97.2%	>95%
Verification Latency	2.1 seconds	<3 sec
Blockchain Throughput	420 tps	≥400 tps
AUC (Fraud Detection)	0.982	>0.95

F. Discussion

The results demonstrate that the integration of blockchain immutability with CNN-driven OCR intelligence significantly enhances the trustworthiness of digital certificates. The blockchain ensures tamper-proof storage and transparent verification, while CNN-OCR provides deep feature-level fraud detection. Compared to existing methods, the proposed system achieves faster verification, higher accuracy, and better scalability under realistic workloads. Table II shows a performance metrics summary. The hybrid design balances

computational efficiency (off-chain OCR processing) with cryptographic assurance (on-chain validation), making it a practical and deployable solution for universities, employers, and certification bodies seeking reliable digital credential management.

VI. CONCLUSION

This paper has introduced a Smart Certificate Validation and Fraud Detection Framework in which blockchain and CNN-based OCR have been combined to ensure safe, automated, and unrestricted certification verification. The system is powerful in dealing with the increasing problem of certificate forgery as it will integrate the impossibility of blockchain with deep learning intelligence in detecting anomalies. The accuracy of identifying forged certificates in experimental results was high with 97.2% and the mean time a verification took 2.1 seconds, which showed that the framework was reliable and scalable. The key contributions of this work are as follows (1) the architecture of a decentralized infrastructure based on blockchain and a transparent certificate management approach, (2) an intelligent CNN-OCR model that can detect visual and textual forgery patterns and (3) the combination of smart contracts and decentralized identities that can support automated and privacy-preserving verification. This framework can be expanded in the future by adding cross-institutional interoperability in the form of federated blockchain networks, improved scaling in Layer-2 solutions and Natural Language Processing (NLP) to validate contextual texts. Also, the modification of the model to the multi-lingual certificates and instant mobile verification might enhance its universal applicability and feasible implementation both in educational and professional spheres.

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